Subject: ASD&D	**Semester:** V
Class: TE	**Batch:** T11
Name: Ananay Amesur	**Roll No:** 2401005

Experiment 1

Aim: Version Control Operations using Git

Lab Outcome: Mapped LO2

LO2: Use Git effectively for version control, source code management and collaborative team development.

Theory:

Git is a distributed version control system that enables developers to track changes in source code, manage multiple versions of a project, and collaborate efficiently with other developers. It allows every developer to maintain a complete copy of the repository, making development faster, reliable, and independent of network connectivity. Version control is an essential practice in modern software engineering as it records every modification made to a project. Git stores project snapshots as commits, each identified by a unique commit hash, thereby maintaining a complete history of the project.

In this project, a local Git repository was created and connected to a remote GitHub repository. Various Git commands were executed to understand the complete version control workflow. Repository status was monitored using git status, project files were staged using git add ., and changes were permanently recorded using git commit. The project history was examined through git log, while differences between file versions were analyzed using git diff. The local repository was synchronized with GitHub using git push, and remote repository details were verified using git remote and git ls-remote origin.

Branch management was performed by creating, switching, and deleting branches to demonstrate isolated feature development without affecting the main branch. Repository cloning was also carried out to understand how an existing remote repository can be copied to another system while preserving its complete commit history and configuration.

The experiment further explored collaboration by adding a collaborator to the GitHub repository, illustrating how multiple developers can contribute to the same project. Rollback

operations were demonstrated using git restore, which discards uncommitted changes and restores files to their previously committed state. Additionally, the git revert command was performed to safely undo a committed change by creating a new commit that reverses the selected commit while preserving the repository history.

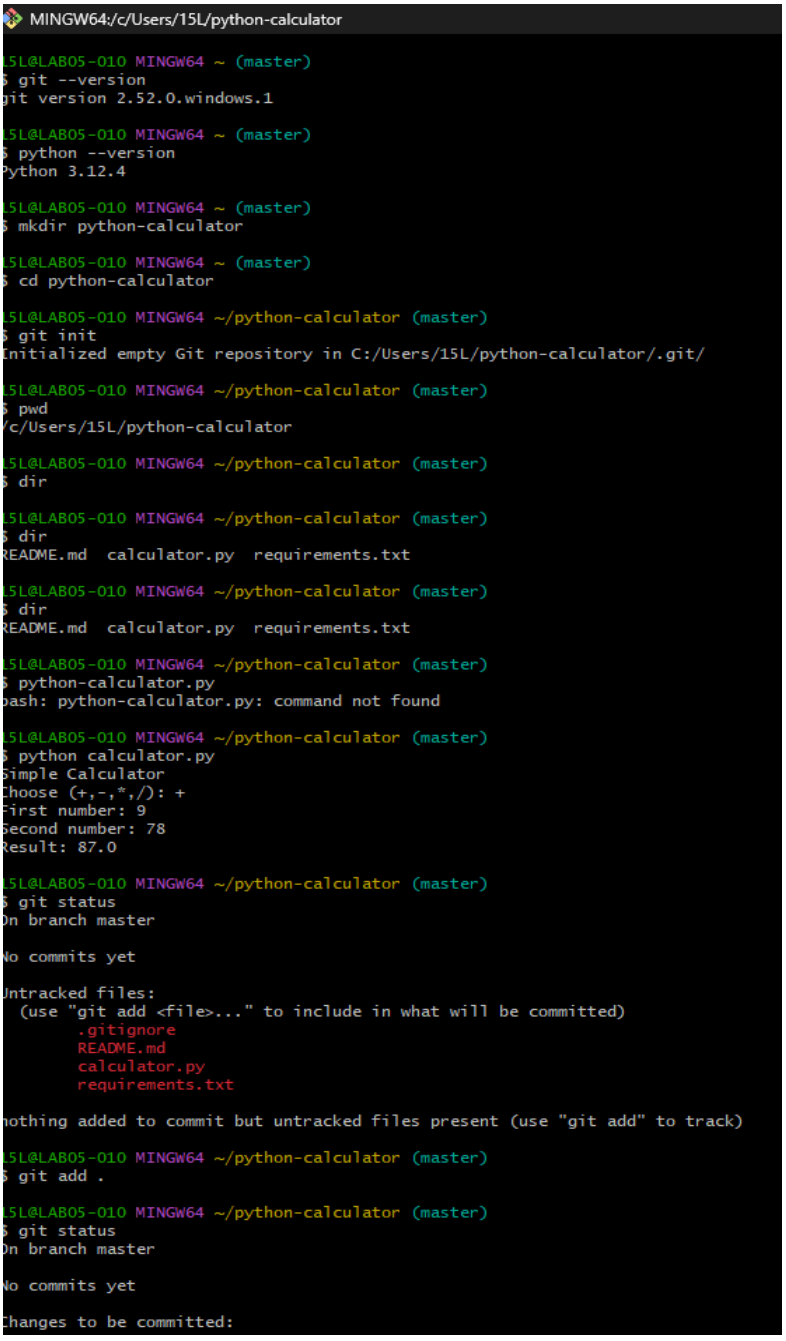
Overall, Git provides an efficient mechanism for source code management by supporting version tracking, branching, collaboration, rollback, and synchronization with remote repositories. These features make Git an indispensable tool for Agile Software Development and DevOps workflows, ensuring code integrity, traceability, and effective team collaboration throughout the software development lifecycle.

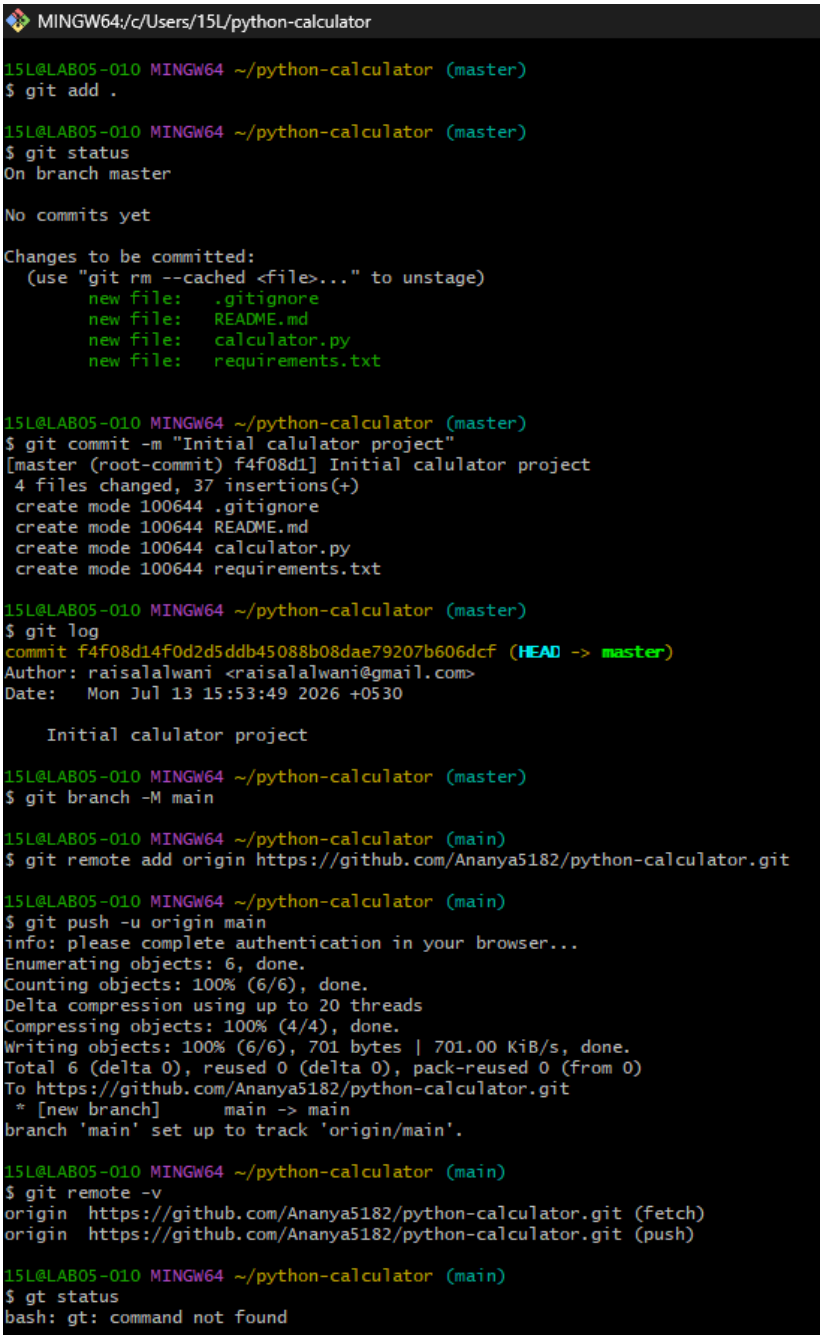
Commands:

Sr. No.	**Git Command**	**Description**
1	dir	Displays the files and folders present in the current working directory.
2	git init	Initializes a new local Git repository in the current directory.
3	git status	Displays the current state of the repository, including modified, staged, and untracked files.
4	git add .	Stages all new and modified files for the next commit.
5	git commit -m "Initial calculator project"	Creates the first snapshot of the project and saves it in the repository history.
6	git commit -m "Added square function"	Records the addition of the square function as a separate version of the project.
7	git commit -m "Added rollback demo"	Commits the changes made to demonstrate rollback and revert operations.
8	git log	Displays the complete commit history with commit hashes, author, date, and messages.
9	git log --oneline	Displays a compact one-line summary of the commit history.
10	git diff	Displays the differences between the working directory and the last committed version of files.
11	git remote add origin <repository-url>	Connects the local repository to a remote GitHub repository.

12	git remote -v	Displays the fetch and push URLs of the configured remote repository.
13	git ls-remote origin	Lists all references (branches and commit hashes) available in the remote repository.
14	git push -u origin main	Uploads the local repository to GitHub and establishes tracking between local and remote branches.
15	git push	Pushes subsequent commits from the local repository to GitHub.
16	git merge feature-square	Merges the changes from the feature-square branch into the main branch
17	git branch	Displays all local branches or creates a new branch when a name is specified.
18	git branch feature-square	Creates a new branch named feature-square without switching to it.
19	git switch feature-square	Switches from the current branch to the feature branch.
20	git switch main	Returns from the feature branch to the main branch.
21	git branch -d feature-square	Deletes the specified branch after it has been merged or is no longer required.
22	git config --list	Displays the Git configuration settings such as username, email, remotes, and repository information.
23	git restore calculator.py	Discards uncommitted changes made to the specified file, restoring its last committed version (rollback).
24	git show	Displays detailed information about the latest commit by default.
25	git clone <repository-url>	Creates a complete local copy of an existing remote Git repository.
26	git pull origin main	Downloads and merges the latest changes from the remote repository into the local repository.
27	git fetch	Downloads updates from the remote repository without merging them into the local branch.
28	git remote show origin	Displays detailed information about the remote repository, including tracked branches and URLs.
29	git config --list	Displays all Git configuration settings currently in use

Output:


MINGW64:/c/Users/15L/python-calculator
15L@LAB05-010 MINGW64 ~ (master)
$ git --version
git version 2.52.0.windows.1
15L@LAB05-010 MINGW64 ~ (master)
$ python --version
Python 3.12.4
15L@LAB05-010 MINGW64 ~ (master)
$ mkdir python-calculator
15L@LAB05-010 MINGW64 ~ (master)
$ cd python-calculator
15L@LAB05-010 MINGW64 ~/python-calculator (master)
$ git init
Initialized empty Git repository in C:/Users/15L/python-calculator/.git/
15L@LAB05-010 MINGW64 ~/python-calculator (master)
$ pwd
/c/Users/15L/python-calculator
15L@LAB05-010 MINGW64 ~/python-calculator (master)
$ dir
15L@LAB05-010 MINGW64 ~/python-calculator (master)
$ dir
README.md calculator.py requirements.txt
15L@LAB05-010 MINGW64 ~/python-calculator (master)
$ dir
README.md calculator.py requirements.txt
15L@LAB05-010 MINGW64 ~/python-calculator (master)
$ python-calculator.py
bash: python-calculator.py: command not found
15L@LAB05-010 MINGW64 ~/python-calculator (master)
$ python calculator.py
Simple Calculator
Choose (+,-,*,/): +
First number: 9
Second number: 78
Result: 87.0
15L@LAB05-010 MINGW64 ~/python-calculator (master)
$ git status
On branch master
No commits yet
Untracked files:
(use "git add <file>..." to include in what will be committed)
.gitignore
README.md
calculator.py
requirements.txt
nothing added to commit but untracked files present (use "git add" to track)
15L@LAB05-010 MINGW64 ~/python-calculator (master)
$ git add .
15L@LAB05-010 MINGW64 ~/python-calculator (master)
$ git status
On branch master
No commits yet
Changes to be committed:


MINGW64:/c/Users/15L/python-calculator
15L@LAB05-010 MINGW64 ~/python-calculator (master)
$ git add .
15L@LAB05-010 MINGW64 ~/python-calculator (master)
$ git status
On branch master
No commits yet
Changes to be committed:
(use "git rm --cached <file>..." to unstage)
new file: .gitignore
new file: README.md
new file: calculator.py
new file: requirements.txt
15L@LAB05-010 MINGW64 ~/python-calculator (master)
$ git commit -m "Initial calulator project"
[master (root-commit) f4f08d1] Initial calulator project
4 files changed, 37 insertions(+)
create mode 100644 .gitignore
create mode 100644 README.md
create mode 100644 calculator.py
create mode 100644 requirements.txt
15L@LAB05-010 MINGW64 ~/python-calculator (master)
$ git log
commit f4f08d14f0d2d5ddb45088b08dae79207b606dcf (HEAD -> master)
Author: raisalalwani <raisalalwani@gmail.com>
Date: Mon Jul 13 15:53:49 2026 +0530
Initial calulator project
15L@LAB05-010 MINGW64 ~/python-calculator (master)
$ git branch -M main
15L@LAB05-010 MINGW64 ~/python-calculator (main)
$ git remote add origin https://github.com/Ananya5182/python-calculator.git
15L@LAB05-010 MINGW64 ~/python-calculator (main)
$ git push -u origin main
info: please complete authentication in your browser...
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 20 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (6/6), 701 bytes | 701.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Ananya5182/python-calculator.git
* [new branch] main -> main
branch 'main' set up to track 'origin/main'.
15L@LAB05-010 MINGW64 ~/python-calculator (main)
$ git remote -v
origin https://github.com/Ananya5182/python-calculator.git (fetch)
origin https://github.com/Ananya5182/python-calculator.git (push)
15L@LAB05-010 MINGW64 ~/python-calculator (main)
$ gt status
bash: gt: command not found

```
MINGW64:/c/Users/15L/python-calculator
$ gt status
bash: gt: command not found

15L@LAB05-010 MINGW64 ~/python-calculator (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   calculator.py

no changes added to commit (use "git add" and/or "git commit -a")

15L@LAB05-010 MINGW64 ~/python-calculator (main)
$ git diff
diff --git a/calculator.py b/calculator.py
index e692683..fff258e 100644
--- a/calculator.py
+++ b/calculator.py
@@ -12,6 +12,9 @@ def divide(a, b):
         return "Cannot divide by zero"
     return a / b

+def square(a):
+    return a*a
+
 print("Simple Calculator")

 choice = input("Choose (+,-,*,/): ")

15L@LAB05-010 MINGW64 ~/python-calculator (main)
$ git add calculator.py

15L@LAB05-010 MINGW64 ~/python-calculator (main)
$ git commit -m "Added square function"
[main df16db1] Added square function
 1 file changed, 3 insertions(+)

15L@LAB05-010 MINGW64 ~/python-calculator (main)
$ git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 20 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 320 bytes | 320.00 KiB/s, done.
Total 3 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/Ananya5182/python-calculator.git
   f4f08d1..df16db1  main -> main

15L@LAB05-010 MINGW64 ~/python-calculator (main)
$ git branch feature-square

15L@LAB05-010 MINGW64 ~/python-calculator (main)
$ git switch feature-square
Switched to branch 'feature-square'

15L@LAB05-010 MINGW64 ~/python-calculator (feature-square)
$ git branch
* feature-square
  main

15L@LAB05-010 MINGW64 ~/python-calculator (feature-square)
$ git add .
```

```
MINGW64:/c/Users/15L/python-calculator
15L@LAB05-010 MINGW64 ~/python-calculator (feature-square)
$ git branch
* feature-square
  main

15L@LAB05-010 MINGW64 ~/python-calculator (feature-square)
$ git add .

15L@LAB05-010 MINGW64 ~/python-calculator (feature-square)
$ git commit -m "Updated square function"
On branch feature-square
nothing to commit, working tree clean

15L@LAB05-010 MINGW64 ~/python-calculator (feature-square)
$ git switch main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.

15L@LAB05-010 MINGW64 ~/python-calculator (main)
$ git merge feature-square
Already up to date.

15L@LAB05-010 MINGW64 ~/python-calculator (main)
$ git push
Everything up-to-date

15L@LAB05-010 MINGW64 ~/python-calculator (main)
$ git branch -d feature-square
Deleted branch feature-square (was df16db1).

15L@LAB05-010 MINGW64 ~/python-calculator (main)
$ git branch
* main

15L@LAB05-010 MINGW64 ~/python-calculator (main)
$ cd /desktop
bash: cd: /desktop: No such file or directory

15L@LAB05-010 MINGW64 ~/python-calculator (main)
$ cd deskto
bash: cd: deskto: No such file or directory

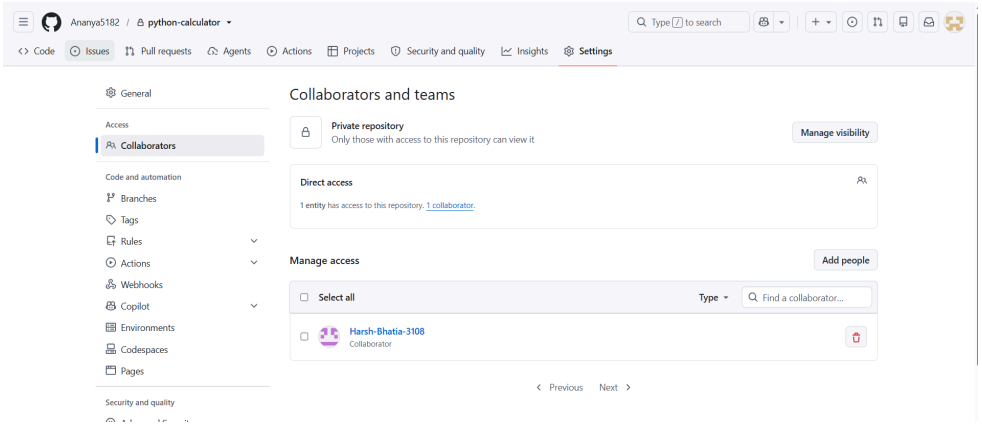
15L@LAB05-010 MINGW64 ~/python-calculator (main)
$ git remote -v
origin  https://github.com/Ananya5182/python-calculator.git (fetch)
origin  https://github.com/Ananya5182/python-calculator.git (push)

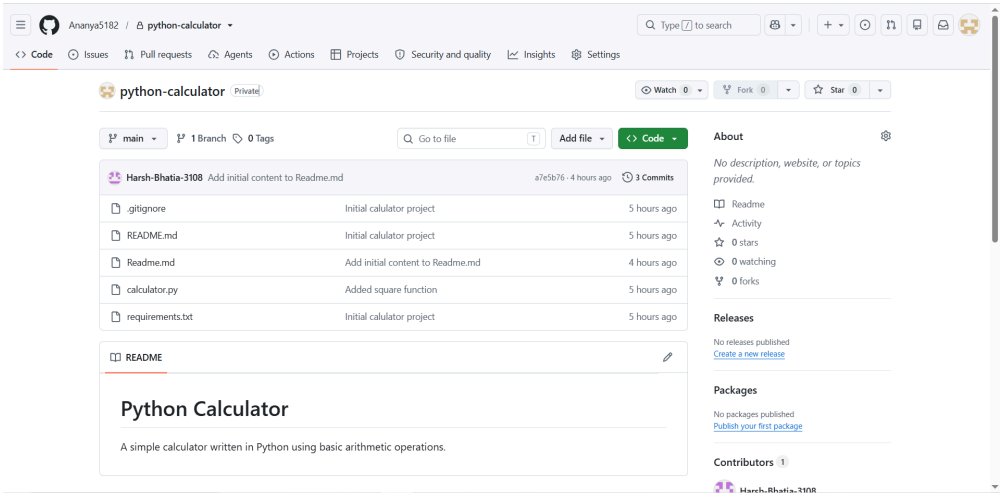
15L@LAB05-010 MINGW64 ~/python-calculator (main)
$ git remote show origin
* remote origin
  Fetch URL: https://github.com/Ananya5182/python-calculator.git
  Push  URL: https://github.com/Ananya5182/python-calculator.git
  HEAD branch: main
  Remote branch:
    main tracked
  Local branch configured for 'git pull':
    main merges with remote main
  Local ref configured for 'git push':
    main pushes to main (up to date)

15L@LAB05-010 MINGW64 ~/python-calculator (main)
$ git ls-remote origin
df16db123cd203a628c53e79c91f71fd42ec3049        HEAD
df16db123cd203a628c53e79c91f71fd42ec3049        refs/heads/main

15L@LAB05-010 MINGW64 ~/python-calculator (main)
$ gh auth status
bash: gh: command not found

15L@LAB05-010 MINGW64 ~/python-calculator (main)
$
```






```
ANANYA@AMESUR MINGW64 ~
$ cd python-calculator

ANANYA@AMESUR MINGW64 ~/python-calculator (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")

ANANYA@AMESUR MINGW64 ~/python-calculator (main)
$ git remote -v
origin  https://github.com/Ananya5182/python-calculator.git (fetch)
origin  https://github.com/Ananya5182/python-calculator.git (push)

ANANYA@AMESUR MINGW64 ~/python-calculator (main)
$ git log --oneline
a7e5b76 (HEAD -> main, origin/main, origin/HEAD) Add initial content to Readme.m
d
df16db1 Added square function
f4f08d1 Initial calulator project

ANANYA@AMESUR MINGW64 ~/python-calculator (main)
$ git pull origin main
From https://github.com/Ananya5182/python-calculator
 * branch            main       -> FETCH_HEAD
Already up to date.

ANANYA@AMESUR MINGW64 ~/python-calculator (main)
$ git fetch

ANANYA@AMESUR MINGW64 ~/python-calculator (main)
$ git remote show origin
* remote origin
  Fetch URL: https://github.com/Ananya5182/python-calculator.git
  Push  URL: https://github.com/Ananya5182/python-calculator.git
  HEAD branch: main
  Remote branch:
    main tracked
  Local branch configured for 'git pull':
    main merges with remote main
  Local ref configured for 'git push':
    main pushes to main (up to date)

ANANYA@AMESUR MINGW64 ~/python-calculator (main)
$ git show
commit a7e5b761b5310a8db99dcc09f8153a15feb4e66a (HEAD -> main, origin/main, orig
in/HEAD)
Author: Harsh-Bhatia-3108 <bhatia.harsh3108@gmail.com>
```

```
ANANYA@AMESUR MINGW64 ~/python-calculator (main)
$ git show
commit a7e5b761b5310a8db99dcc09f8153a15feb4e66a (HEAD -> main, origin/main, orig
in/HEAD)
Author: Harsh-Bhatia-3108 <bhatia.harsh3108@gmail.com>
Date:   Mon Jul 13 16:49:54 2026 +0530

    Add initial content to Readme.md

diff --git a/Readme.md b/Readme.md
new file mode 100644
index 0000000..dd2517c
--- /dev/null
+++ b/Readme.md
@@ -0,0 +1 @@
+hiiii

ANANYA@AMESUR MINGW64 ~/python-calculator (main)
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=schannel
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
filter.lfs.clean=git-lfs clean -- %f
user.name=Ananya5182
user.email=ananyaamesur@gmail.com
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
core.symlinks=false
core.ignorecase=true
remote.origin.url=https://github.com/Ananya5182/python-calculator.git
remote.origin.fetch=+refs/heads/*:refs/remotes/origin/*
branch.main.remote=origin
branch.main.merge=refs/heads/main

ANANYA@AMESUR MINGW64 ~/python-calculator (main)
$ git branch
* main
```

```
ANANYA@AMESUR MINGW64 ~/python-calculator (main)
$ git log --graph --all
* commit a7e5b761b5310a8db99dcc09f8153a15feb4e66a (HEAD -> main, origin/main, or
igin/HEAD)
| Author: Harsh-Bhatia-3108 <bhatia.harsh3108@gmail.com>
| Date:   Mon Jul 13 16:49:54 2026 +0530
|
|     Add initial content to Readme.md
|
* commit df16db123cd203a628c53e79c91f71fd42ec3049
| Author: raisalalwani <raisalalwani@gmail.com>
| Date:   Mon Jul 13 16:12:18 2026 +0530
|
|     Added square function
|
* commit f4f08d14f0d2d5ddb45088b08dae79207b606dcf
  Author: raisalalwani <raisalalwani@gmail.com>
  Date:   Mon Jul 13 15:53:49 2026 +0530

      Initial calulator project

ANANYA@AMESUR MINGW64 ~/python-calculator (main)
$ git stash
Saved working directory and index state WIP on main: a7e5b76 Add initial content
 to Readme.md

ANANYA@AMESUR MINGW64 ~/python-calculator (main)
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=schannel
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
filter.lfs.clean=git-lfs clean -- %f
user.name=Ananya5182
user.email=ananyaamesur@gmail.com
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
```

MINGW64:/c/Users/ANANYA

```
ANANYA@AMESUR MINGW64 ~/python-calculator (main)
$ cd ~/Documents

ANANYA@AMESUR MINGW64 ~/Documents
$ pwd
/c/Users/ANANYA/Documents

ANANYA@AMESUR MINGW64 ~/Documents
$ git clone https://github.com/Ananaya5182/python-calculator.git
Cloning into 'python-calculator'...
remote: Repository not found.
fatal: repository 'https://github.com/Ananaya5182/python-calculator.git/' not fo
und

ANANYA@AMESUR MINGW64 ~/Documents
$ git clone https://github.com/Ananya5182/python-calculator.git
Cloning into 'python-calculator'...
remote: Enumerating objects: 12, done.
remote: Counting objects: 100% (12/12), done.
remote: Compressing objects: 100% (7/7), done.
remote: Total 12 (delta 3), reused 8 (delta 2), pack-reused 0 (from 0)
Receiving objects: 100% (12/12), done.
Resolving deltas: 100% (3/3), done.
warning: the following paths have collided (e.g. case-sensitive paths
on a case-insensitive filesystem) and only one from the same
colliding group is in the working tree:

  'README.md'
  'Readme.md'

ANANYA@AMESUR MINGW64 ~/Documents
$ cd python-calculator

ANANYA@AMESUR MINGW64 ~/Documents/python-calculator (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")

ANANYA@AMESUR MINGW64 ~/Documents/python-calculator (main)
$ git remote -v
origin  https://github.com/Ananya5182/python-calculator.git (fetch)
origin  https://github.com/Ananya5182/python-calculator.git (push)

ANANYA@AMESUR MINGW64 ~/Documents/python-calculator (main)
$ cd ..

ANANYA@AMESUR MINGW64 ~/Documents
$ cd ..

ANANYA@AMESUR MINGW64 ~
```

```
ANANYA@AMESUR MINGW64 ~
$ cd python-calculator

ANANYA@AMESUR MINGW64 ~/python-calculator (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   Readme.md

no changes added to commit (use "git add" and/or "git commit -a")

ANANYA@AMESUR MINGW64 ~/python-calculator (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   Readme.md
        modified:   calculator.py

no changes added to commit (use "git add" and/or "git commit -a")

ANANYA@AMESUR MINGW64 ~/python-calculator (main)
$ ^C

ANANYA@AMESUR MINGW64 ~/python-calculator (main)
$ git restore calculator.py

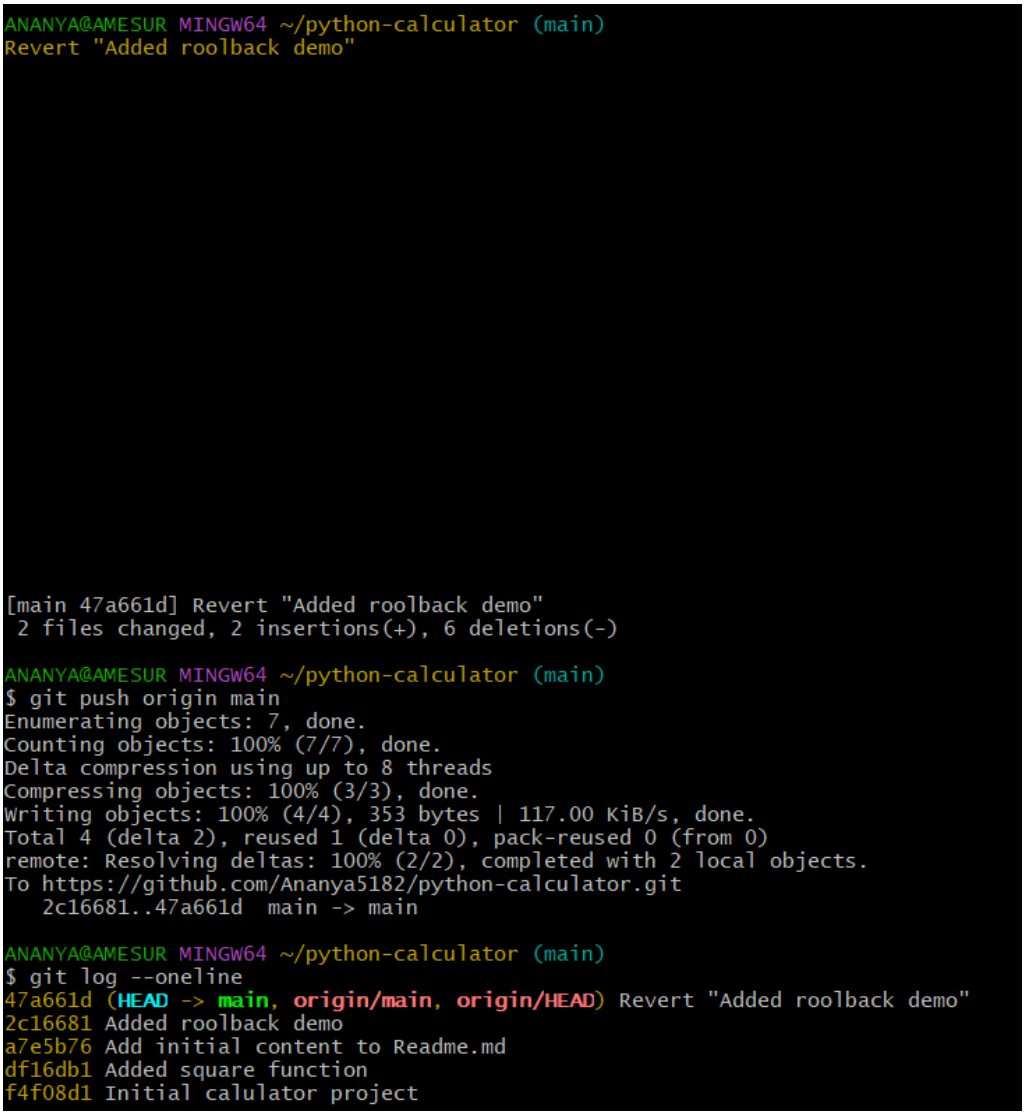
ANANYA@AMESUR MINGW64 ~/python-calculator (main)
$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   Readme.md

no changes added to commit (use "git add" and/or "git commit -a")

ANANYA@AMESUR MINGW64 ~/python-calculator (main)
$ git add .

ANANYA@AMESUR MINGW64 ~/python-calculator (main)
$ git commit -m "Added roolback demo"
[main 2c16681] Added roolback demo
 2 files changed, 6 insertions(+), 2 deletions(-)
```



Conclusion:

The experiment successfully demonstrated the fundamental operations of Git as a distributed version control system used in Agile Software Development and DevOps. Various Git commands such as git status, git add, git commit, git log, git diff, git push, git clone, git remote, git ls-remote, and git config were executed to understand how Git tracks, stores, and synchronizes project changes.

This experiment was instrumental in helping gain an understanding of Git's role in maintaining project integrity, facilitating collaborative development, and supporting the version control practices essential in Agile Software Development and DevOps environments. The experiment improved my real-world understanding of Git and its significance in modern software engineering workflows.